A maximum parsimony-based method for reconstructing multi-chromosomal rearrangements with copy number changes

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Abstract

This document is used for formalizing a framework for reconstructing a multi-chromosomal genome B from genome A using a sequence of (k, d, g)-break operations that combine rearrangements, deletions and duplications (gains), under the infinite sites assumption (ISA). Enable enumeration and optimization of such scenarios using the extended breakpoint graph model.

1 Introduction

1.1 General definitions in genome representations

A **synteny block** b is an oriented sequence of base pairs. Each block has two **extremities**: tail t_b and head h_b ($b = \{t_b, h_b\}$). A **genome** is a collection of chromosomes, where each **chromosome** is a sequence of signed synteny blocks. Chromosomes may be linear or circular. An **adjacency** is an unordered pair of extremities (e.g., $\{h_i, t_j\}$). Valid adjacencies occur either between extremities of adjacent blocks or between an extremity and a telomere node. **Telomere** nodes are vertices in the graph and are used to model the ends of linear chromosomes. A linear chromosome has two telomere nodes: T_i, the tail-side telomere (connected to t_{b_1}, the first block), and H_i, the head-side telomere (connected to h_{b_n}, the last block). A circular chromosome has no telomere nodes.

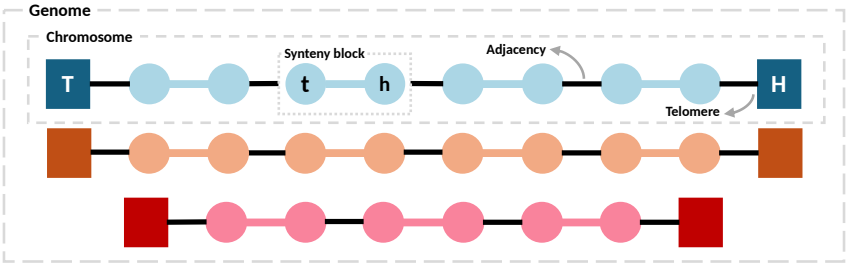


Figure 1: Genome representation using synteny blocks and adjacencies. Each chromosome is a set of signed synteny blocks, and each block b is represented by two extremities, tail t_b and head h_b. Black edges between extremities indicate adjacencies between consecutive blocks, while telomere nodes T_i and H_i mark the ends of chromosomes

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1.2 k-break

A **k-break** is a genome rearrangement operation that models the simultaneous cutting of k adjacencies and the subsequent rejoining of their $2k$ extremities into k new adjacencies. Formally, let $\tau = (\alpha, \beta)$ be a k-break, such that $\alpha = \{\{u_1, u_2\}, \ldots, \{u_{2k-1}, u_{2k}\}\}$ is a set of k adjacencies in genome A, and let $U = \{u_1, \ldots, u_{2k}\}$ be the set of free extremities introduced by cutting α. Let σ be a pairing of the elements of U, and let β be the set of k new adjacencies induced by this pairing:

$$\beta = \big\{\{u, v\} : u, v \in U \text{ and } u \text{ is paired with } v \text{ by } \sigma\big\}.$$

The operation τ removes the adjacencies in α and replaces them with the adjacencies in β. The resulting genome differs from A only in its adjacencies, no new synteny blocks are introduced or removed. Figure 2A demonstrates a 4-break operation.

1.3 Breakpoint graph

A breakpoint graph (BG) provides a graphical representation of the differences in adjacency structure between two genomes. Let A and B be two genomes containing the same set of synteny blocks, and let V be the set of all block extremities and telomere vertices. For each genome $X \in \{A, B\}$, let E_X denote the set of edges induced by its adjacencies:

$$E_X = \big\{\{u, v\} : \{u, v\} \text{ is an adjacency in genome } X\big\}.$$

Define

$$E_A^{\Delta} = E_A \setminus E_B \qquad \text{and} \qquad E_B^{\Delta} = E_B \setminus E_A.$$

The BG of A and B is the two-edge-colored graph

$$BG(A, B) = \big(V, E_A^{\Delta} \cup E_B^{\Delta}\big),$$

where the edges in E_A^{Δ} and E_B^{Δ} are assigned distinct colors. Thus, adjacencies shared by A and B are omitted from the BG, whereas an adjacency present in only one of the genomes is represented by an edge of the corresponding color.

Under the present representation, each genome induces a perfect matching on V. Consequently, every non-isolated vertex of $BG(A, B)$ is incident to exactly one edge corresponding to an adjacency in A and exactly one edge corresponding to an adjacency in B. The edges therefore alternate between the two colors, and each nontrivial connected component of $BG(A, B)$ is an even alternating cycle. Isolated vertices correspond to extremities whose adjacencies are identical in the two genomes.

A k-break $\tau = (\alpha, \beta)$ applied to genome A replaces the adjacencies in α with those in β. If $A' = \tau(A)$, then

$$E_{A'} = (E_A \setminus \alpha) \cup \beta.$$

In the particular case where τ directly transforms A into B, the removed and introduced adjacencies satisfy

$$\alpha = E_A \setminus E_B \qquad \text{and} \qquad \beta = E_B \setminus E_A,$$

provided that the operation contains no redundant cuts. The BG then consists precisely of the adjacencies removed by the operation and the new adjacencies introduced in their place. More generally, transforming A into B can be viewed as a sequence of k-breaks that modifies the A-colored edges until no nonshared adjacency remains, at which point $E_A = E_B$ and the BG contains only isolated vertices. Figure 2B presents the BG for the two genomes shown in Figure 2A.

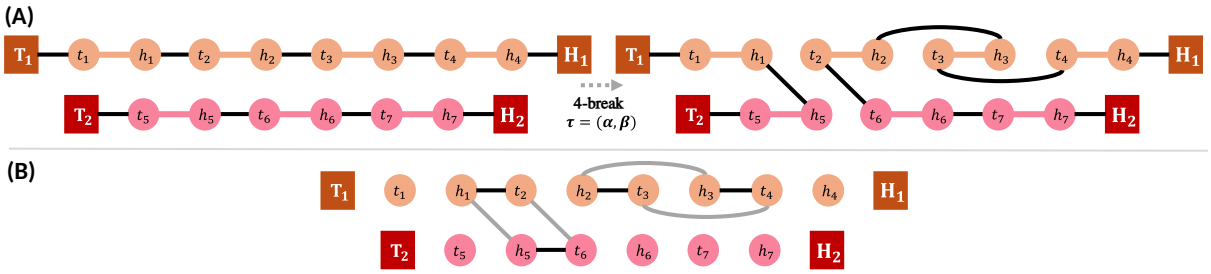


Figure 2: (A) Illustration of a 4-break $\tau = (\alpha, \beta)$, where $\alpha = \{\{h_1, t_2\}, \{h_2, t_3\}, \{h_3, t_4\}, \{h_5, t_6\}\}$ is the set of four adjacencies removed from the original genome, and $\beta = \{\{h_1, h_5\}, \{t_2, t_6\}, \{h_2, h_3\}, \{t_3, t_4\}\}$ is the set of four new adjacencies induced by the new pairing σ. (B) The corresponding BG $BG(A, \tau(A))$, in which the black edges represent the removed adjacencies in α and the gray edges represent the introduced adjacencies in β. Adjacencies shared by the two genomes are omitted, leaving two alternating cycles; all remaining extremities and telomere vertices are isolated.

2 Methods

2.1 (k, d)-break

In this classical setting, k-breaks are typically defined under the assumption that the two genomes being compared - A and B - contain the same set of synteny blocks, possibly reordered and reoriented. However, this assumption does not hold in many biological contexts, such as cancer or germline rearrangements, where blocks may be deleted entirely during evolution. The (k, d)-**break** extends this model by allowing the deletion of up to d contiguous intervals of synteny blocks (and their adjacencies) during a single event.

Formally, we consider a set of k adjacencies $\alpha = \{\{u_1, u_2\}, \ldots, \{u_{2k-1}, u_{2k}\}\}$ to be removed. The resulting set of $2k$ exposed extremities, denoted $U = \bigcup \alpha$, is then reconnected in one of two ways: (1) paired together into new adjacencies, as in a classical k-break, or (2) connected to a special structure representing a deleted interval. Specifically, for each deleted interval, a pair of artificial nodes is introduced, denoted $T^{(i)}_{\text{DC}}, H^{(i)}_{\text{DC}}$ for $i = 1, \ldots, d$, and the two terminal extremities of that interval are connected to these "deletion collector" (DC) nodes. Each DC pair initially forms an adjacency between $T^{(i)}_{\text{DC}}$ and $H^{(i)}_{\text{DC}}$, which is broken during the event and replaced by the connections to the exposed extremities, thereby modeling excision (deletion) of the synteny block.

Definition 1. A (k, d)-**break** is a triple $\tau = (\alpha, \beta, \delta)$, where:

- $\alpha = \{\{u_1, u_2\}, \ldots, \{u_{2k-1}, u_{2k}\}\}$ is the set of k adjacencies to be removed.
- $\delta = \{\{x_1, y_1\}, \ldots, \{x_r, y_r\}\}$ is a set of $r \leq d$ unordered pairs of exposed extremities, each representing the two ends of a deleted block.
- Let $U = \bigcup \alpha$, and let $\mathcal{D} = \bigcup \delta$. Let σ be a pairing of the remaining extremities $U \setminus \mathcal{D}$, and let β be the set of new adjacencies induced by this pairing:

$$\beta = \big\{\{u, v\} : u, v \in U \setminus \mathcal{D} \text{ and } u \text{ is paired with } v \text{ by } \sigma\big\}.$$

  Thus, β is a perfect matching on the remaining $2k - 2r$ extremities.

Each pair $\{x, y\} \in \delta$ is associated with a unique pair of DC telomeres $\{T^{(i)}_{\text{DC}}, H^{(i)}_{\text{DC}}\}$, which are connected by an adjacency before the operation. During the (k, d)-break, their internal adjacency is removed, and the two extremities x and y are connected to $T^{(i)}_{\text{DC}}$ and $H^{(i)}_{\text{DC}}$, respectively. The DC pair thus forms a synthetic linear chromosome representing the excised segment.

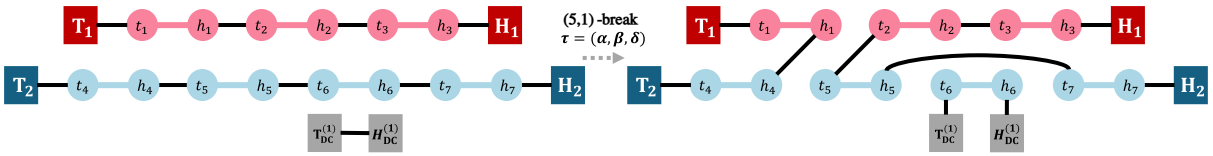


Figure 3: Illustration of a $(5,1)$-break $\tau = (\alpha, \beta, \delta)$, where $\alpha = \{\{h_1, t_2\}, \{h_4, t_5\}, \{h_5, t_6\}, \{h_6, t_7\}, \{T_{DC}^{(1)}, H_{DC}^{(1)}\}\}$ is the set of five adjacencies removed, $\delta = \{\{t_6, h_6\}\}$ specifies the extremities of the deleted block, and $\beta = \{\{h_1, h_5\}, \{t_2, h_4\}, \{t_5, t_7\}\}$ pairs the remaining exposed extremities; the deleted extremities t_6 and h_6 are connected to $T_{\mathrm{DC}}^{(1)}$ and $H_{\mathrm{DC}}^{(1)}$, respectively.

This model allows us to express complex rearrangement events, including those involving inversion, translocation, and deletion, in a unified algebraic framework. When $d = 0$, the model reduces to the classical k-break model; when $d > 0$, the artificial chromosomes flanked by DC nodes represent excised fragments.

2.2 (k, d, g)-break

The (k, d)-break model allows deletions during a k-break event, but assumes no new block copies are introduced. In many biological settings, however, new copies of existing blocks can be created and inserted into novel positions, sometimes multiple different blocks in a single event. The (k, d, g)-**break** model extends the (k, d)-break model by allowing up to g such insertions (*gains*), drawn from a set of latent block copies that are already represented in an augmented genome A^+ ($A \subseteq A^+$). Each block b in A has extra copies b^+ stored on its own artificial chromosome capped by a unique "addition-holder" (AH) telomere pair $(T_{\mathrm{AH}}^{(b)}, H_{\mathrm{AH}}^{(b)})$ with adjacencies

$$\{t_b^+, T_{\mathrm{AH}}^{(b)}\}, \quad \{H_{\mathrm{AH}}^{(b)}, h_b^+\}.$$

These AH adjacencies are part of A^+ but absent from B, and can be broken if the extra copy b^+ is to be inserted.

Definition 2. A (k, d, g)-**break** is a quadruple $\tau = (\alpha, \beta, \gamma, \delta)$ on A^+, where:

- $\alpha = \{\{u_1, u_2\}, \ldots, \{u_{2k-1}, u_{2k}\}\}$ is the set of k A^+-adjacencies to be removed. These may include ordinary adjacencies of A (potential insertion points), DC pair edges $\{T_{\mathrm{DC}}^{(i)}, H_{\mathrm{DC}}^{(i)}\}$, AH attachment edges $\{t_b^+, T_{\mathrm{AH}}^{(b)}\}$ ans $\{H_{\mathrm{AH}}^{(b)}, h_b^+\}$.
- $\delta = \{\{x_1, y_1\}, \ldots, \{x_r, y_r\}\}$ is a set of $r \leq d$ unordered pairs of exposed extremities representing the two ends of contiguous deleted intervals, with the property that for each $\{x_i, y_i\} \in \delta$, the corresponding DC pair $\{T_{\mathrm{DC}}^{(i)}, H_{\mathrm{DC}}^{(i)}\} \in \alpha$.
- $\gamma = \{\{t_j^+, h_j^+\} \mid j = 1, \ldots, s\}$ is the set of $s \leq g$ unordered pairs of extremities that correspond to the additional copies b_j^+ of the blocks to be inserted. For each $\{t_j^+, h_j^+\} \in \gamma$, both AH attachment edges $\{t_j^+, T_{\mathrm{AH}}^{(j)}\} \in \alpha, \{H_{\mathrm{AH}}^{(j)}, h_j^+\} \in \alpha$ are cut, making b_j^+ available for integration.
- β is a perfect matching on the set of exposed extremities $U = \bigcup \alpha$ subject to:
  1. For each $\{x_i, y_i\} \in \delta$, $\{x_i, T_{\mathrm{DC}}^{(i)}\}, \{H_{\mathrm{DC}}^{(i)}, y_i\} \in \beta$.
  2. For each $\{t_j^+, h_j^+\} \in \gamma$, $\{T_{\mathrm{AH}}^{(j)}, H_{\mathrm{AH}}^{(j)}\} \in \beta$.
  3. All remaining extremities in $U \setminus \mathcal{D}$ are paired in β, where $\mathcal{D} = \bigcup \delta$ $(\mathcal{A} = \bigcup \gamma, \mathcal{A} \subset U)$.

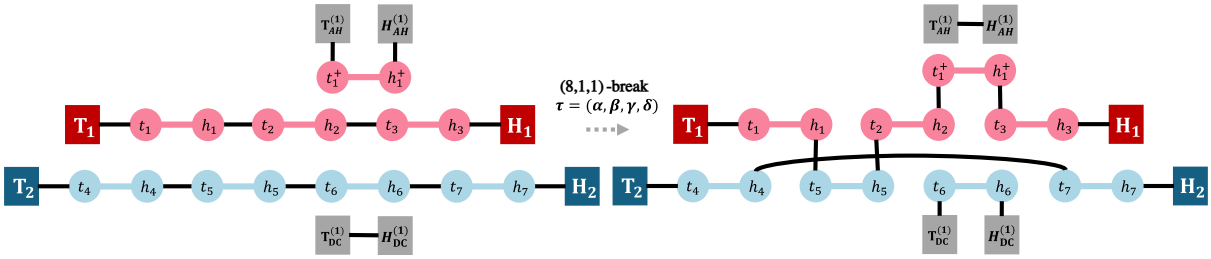


Figure 4: Illustration of an $(8,1,1)$-break $\tau = (\alpha, \beta, \gamma, \delta)$, where $\alpha = \{\{h_1, t_2\}, \{h_2, t_3\}, \{h_4, t_5\}, \{h_5, t_6\}, \{h_6, t_7\}, \{t_1^+, T_{\text{AH}}^{(1)}\}, \{h_1^+, H_{\text{AH}}^{(1)}\}, \{T_{\text{DC}}^{(1)}, H_{\text{DC}}^{(1)}\}\}$ is the set of eight removed adjacencies, $\gamma = \{\{t_1^+, h_1^+\}\}$ specifies the additional copy of block b_1 to be inserted, $\delta = \{\{t_6, h_6\}\}$ specifies the deleted block b_6, and $\beta = \{\{h_1, t_5\}, \{t_2, h_5\}, \{h_4, t_7\}, \{h_2, t_1^+\}, \{h_1^+, t_3\}, \{t_6, T_{\text{DC}}^{(1)}\}, \{h_6, H_{\text{DC}}^{(1)}\}, \{T_{\text{AH}}^{(1)}, H_{\text{AH}}^{(1)}\}\}$ is the resulting perfect matching, which inserts b_1^+, excises b_6, and reconnects the remaining exposed extremities.

The perfect matching β determines both the insertion points and the ordering/orientation of inserted blocks. When multiple elements of γ are consecutively connected between two host extremities in β, the corresponding blocks are inserted adjacently in that order, yielding a continuous addition.

Definition 3. Let A and B be genomes. A (k, d, g)**-scenario** transforming A into B is a finite sequence of (k, d, g)-breaks:

$$T = (\tau_1, \tau_2, \ldots, \tau_\ell), \quad \tau_i = (\alpha_i, \beta_i, \gamma_i, \delta_i),$$

such that τ_1 is applied to genome A, and each subsequent τ_{i+1} is applied to the components of the genome resulting from τ_i. The final step τ_ℓ produces a genome whose set of adjacencies corresponds to B. The scenario T encodes both the rearranged adjacencies and any segments deleted or gained along the transformation from A to B.

This model expresses complex rearrangement events, including inversions, translocations, deletions, duplications, and tandem additions, in a unified algebraic framework. It lifts the constraint that genomes must contain the same set of blocks, enabling representation of scenarios where certain blocks are completely lost and others are duplicated in new positions. When $d = g = 0$, the model reduces to the classical k-break; when $d > 0$ or $g > 0$, the DC and AH nodes explicitly represent excised and inserted fragments, respectively. This formulation sets the foundation for graph-based representations and scenario enumeration that incorporate both rearrangement, loss, and gain.

2.3 Extended breakpoint graph

Given two genomes A and B, the **breakpoint graph** $G(A, B)$ is a 2-edge-colored graph constructed over a set of vertices corresponding to the extremities of all synteny blocks, as well as any telomere nodes used to terminate linear chromosomes. Edges in the breakpoint graph are classified as follows:

- **Black edges** represent adjacencies present in genome A but not in genome B.
- **Gray edges** represent adjacencies present in genome B but not in genome A.

This classical structure captures copy-neutral rearrangements but does not represent deletions or gains explicitly. To handle both loss and gain events introduced by (k, d, g)-breaks, we define an extended version of the breakpoint graph.

Definition 4. Given two genomes A and B, the **extended breakpoint graph** (EBG) augments the classical breakpoint graph with additional vertices and edges that represent both deleted and newly gained intervals. Specifically:

- For each deleted interval, as specified by a pair of extremities in δ from some (k, d, g)-break, we introduce a unique pair of DC telomeres, denoted $T_{\text{DC}}^{(i)}$ and $H_{\text{DC}}^{(i)}$, for $i = 1, \ldots, |\delta|$. These DC nodes are treated as artificial telomeres that form linear chromosomes. The graph includes an initial adjacency $\{T_{\text{DC}}^{(i)}, H_{\text{DC}}^{(i)}\}$ for each such pair.
- For each gained block copy b^+, as specified by a pair of extremities in γ from some (k, d, g)-break, we introduce a unique pair of AH telomeres, denoted $T_{\text{AH}}^{(b)}$ and $H_{\text{AH}}^{(b)}$. These AH nodes initially cap the extra copy b^+ as a separate artificial chromosome with adjacencies $\{T_{\text{AH}}^{(b)}, t_b^+\}$ and $\{h_b^+, H_{\text{AH}}^{(b)}\}$.

Edges in the EBG are constructed as follows:

- **Black edges**:
  - Adjacencies in genome A but not in B.
  - The DC pair edges $\{T_{\text{DC}}^{(i)}, H_{\text{DC}}^{(i)}\}$ before deletion (present in A but absent from B).
  - The AH attachment edges $\{T_{\text{AH}}^{(b)}, t_b^+\}$ and $\{h_b^+, H_{\text{AH}}^{(b)}\}$ for gains (present in A^+ but absent from B).
- **Gray edges**:
  - Adjacencies in genome B but not in A.
  - For each deletion $\{x_i, y_i\} \in \delta$, edges $\{x_i, T_{\text{DC}}^{(i)}\}$ and $\{y_i, H_{\text{DC}}^{(i)}\}$ linking the deleted extremities to their DC pair (present in B but absent from A).
  - Edges connecting the inserted copy b^+ into its host location, as specified by the perfect matching β. If multiple gained blocks are inserted adjacently as a tandem gain, additional edges between their copies (e.g., $\{h_{b_1}^+, t_{b_2}^+\}$) appear as gray edges, representing internal adjacencies of the continuous insertion.

The EBG thus generalizes the classical breakpoint graph to support explicit modeling of both structural rearrangements and content changes, including deletions and gains. By representing both DC and AH nodes, it enables the analysis of (k, d, g)-scenarios that alter the order, orientation, and copy number of synteny blocks.

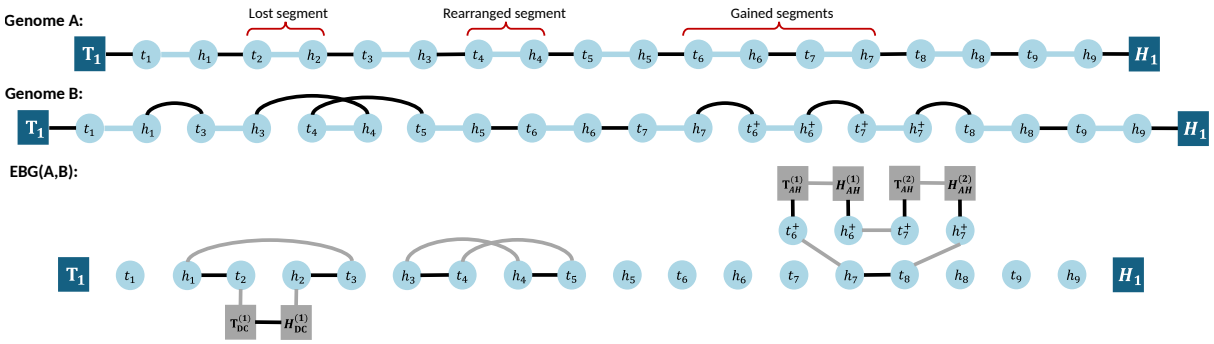


Figure 5: Illustration of the extended breakpoint graph $EBG(A, B)$. Genome B is obtained from genome A by deleting block b_2, inverting block b_4, and inserting additional copies b_6^+ and b_7^+ as a consecutive gain between blocks b_7 and b_8. The upper rows show genomes A and B, while the lower row shows their extended breakpoint graph. Black edges represent adjacencies present on the A but absent from B, including the DC-pair adjacency $\{T_{\mathrm{DC}}^{(1)}, H_{\mathrm{DC}}^{(1)}\}$ and the AH attachment edges that cap the latent copies b_6^+ and b_7^+. Gray edges represent adjacencies present in B but absent from A, including the adjacency $\{h_1, t_3\}$ created by the deletion, the edges connecting t_2 and h_2 to their DC telomeres, the adjacencies induced by the inversion of b_4, and the edges that integrate b_6^+ and b_7^+ into the host chromosome. Adjacencies shared by the two genomes are omitted, and isolated vertices correspond to unchanged extremities.

2.4 EBG under the infinite sites assumption

Definition 5 (ISA for (k, d, g)-scenarios)**.** A (k, d, g)-scenario T satisfies the *infinite sites assumption* (ISA) if each extremity participates in at most one breakpoint over the entire scenario. Formally, no extremity appears more than once in the union of all α_i, β_i, γ_i and δ_i across all steps of T. This implies that breakpoint reuse does not occur.

A CN gain introduces a new copy of a block, while leaving the original copy unchanged. The original block retains its original adjacencies, and the newly gained copy has its own distinct extremities. Thus, the rearrangement acts on the extremities of the new copy rather than reusing those of the original block, so the ISA remains valid.

Technically, in the (k, d, g)-model, each spare copy b^+ used in a gain already exists in the augmented starting genome A^+ as an isolated chromosome capped by its AH pair. A gain removes only the AH attachments of b^+, leaving the original copy b and its adjacencies untouched. Consequently, even if b is deleted or moved later, the extremities involved in the gain and the deletion are distinct, so the ISA is preserved.

Lemma 1 (**structure under ISA**)**.** *Let A and B be genomes such that B is obtained from A through a (k, d, g)-scenario that satisfies the ISA, and let $G(A, B)$ denote the EBG. Then every connected component of $G(A, B)$ is a color-alternating cycle or a single isolated vertex.*

Proof of Lemma 1. Let $G(A, B)$ be the EBG obtained from genomes A and B. Black edges correspond to adjacencies in $A \setminus B$ and gray edges to adjacencies in $B \setminus A$.

Vertex degrees. Fix a vertex u of $G(A, B)$, either a block extremity, a telomere, a DC, or an AH associated with a spare copy used in a gain.

(a) If the same adjacency containing u appears in both A and B, the corresponding edge is common to both genomes and is therefore omitted from the $G(A, B)$; u has degree 0.

(b) If u is a block extremity that lies in an adjacency of A that is absent from B, one *black* edge is incident to u. Under ISA, u participates in *exactly one* adjacency of B; that adjacency differs from the black one, therefore a single *gray* edge is also incident to u. The node has degree 2.

(c) For a DC node $T_{\mathrm{DC}}^{(i)}$ (resp. $H_{\mathrm{DC}}^{(i)}$) the pre-deletion adjacency $\{T_{\mathrm{DC}}^{(i)}, H_{\mathrm{DC}}^{(i)}\}$ is present only in A and contributes one *black* edge, whereas the post-deletion adjacency $\{x_i, T_{\mathrm{DC}}^{(i)}\}$ (resp. $\{y_i, H_{\mathrm{DC}}^{(i)}\}$) is present only in B and contributes one *gray* edge; the node again has degree 2.

(d) For an AH node $T_{\mathrm{AH}}^{(b)}$ (resp. $H_{\mathrm{AH}}^{(b)}$) corresponding to a spare copy b^+ used in a gain, the pre-gain attachment $\{T_{\mathrm{AH}}^{(b)}, t_{b^+}\}$ (resp. $\{h_{b^+}, H_{\mathrm{AH}}^{(b)}\}$) is present in A^+ but absent from B and contributes one black edge. The post-gain adjacency $\{T_{\mathrm{AH}}^{(b)}, H_{\mathrm{AH}}^{(b)}\}$ is present only in B and contributes one gray edge. Thus, each AH node has degree 2.

(e) For the extremities of the gained copy t_{b^+} and h_{b^+}, the pre-gain attachments to their AH nodes are black, while the post-gain attachments connect them into their host location via the perfect matching β (or, in the case of a tandem gain, to the extremity of another gained block), which are gray. Thus, each extremity of a gained copy also has degree 2.

Hence every vertex of the EBG, whether a block extremity, telomere, DC node, or AH node, has degree either 0 (no incident edge) or 2, with exactly one black and one gray edge.

Connected components. $G(A, B)$ is a finite graph in which all vertices have degree 0 or 2, thus the EBG decomposes into isolated vertices (components of size one and degree 0), and cycles (2-regular subgraphs of size ≥ 2).

In the latter case, incident edges at every vertex are of opposite colors, so each cycle is color-alternating.

Conclusion. Every connected component of $G(A, B)$ is either a single isolated vertex or a color-alternating cycle, as required. □

Lemma 2 (**cycle-to-(k, d, g)-break mapping**). *Let A and B be genomes such that B is obtained from A through a (k, d, g)-scenario satisfying the ISA, and let $G(A, B)$ be the EBG with alternating cycles $C = \{C_1, \ldots, C_m\}$. Each C_i defines a (k_i, d_i, g_i)-break $\tau_i = (\alpha_i, \beta_i, \gamma_i, \delta_i)$, where α_i and β_i are the sets of black and gray edges in C_i, δ_i specifies the deletion intervals represented in the cycle through connections to DC telomeres, and γ_i specifies the pairs of extremities corresponding to the extra copies of blocks to be inserted.*

Proof of Lemma 2. We prove the lemma by constructing a (k, d, g)-break for each cycle in the EBG.

Construction (by cycle). Let $G(A, B)$ be the EBG under ISA with alternating cycles $C = \{C_1, \ldots, C_m\}$. Fix a cycle C_i and set $\alpha_i = \{\text{black edges of } C_i\}$, $\beta_i = \{\text{gray edges of } C_i\}$. Let δ_i collect each unordered pair $\{x, y\}$ of block extremities whose gray edges in C_i connect x to $T_{\mathrm{DC}}^{(p)}$ and y to $H_{\mathrm{DC}}^{(p)}$ for the same DC pair p. Let γ_i collect each unordered pair $\{t_b^+, h_b^+\}$ of extremities of an extra block copy whose black edges in C_i connect t_b^+ to $T_{\mathrm{AH}}^{(b)}$ and h_b^+ to $H_{\mathrm{AH}}^{(b)}$ for the same AH pair. Define $\tau_i = (\alpha_i, \beta_i, \gamma_i, \delta_i)$.

Verification that τ_i is a (k_i, d_i, g_i)-break. From Lemma 1, in a cycle under ISA, every vertex has exactly one black and one gray incident edge; hence the black edges in C_i are pairwise vertex-disjoint. Cutting α_i removes $k_i = |\alpha_i|$ adjacencies of A and exposes precisely $U_i = \bigcup \alpha_i$. If a DC pair $\{T_{\mathrm{DC}}^{(p)}, H_{\mathrm{DC}}^{(p)}\}$ occurs in C_i, its black edge lies in α_i and is also cut, exposing those telomeres. Similarly, if an AH pair $\{T_{\mathrm{AH}}^{(b)}, H_{\mathrm{AH}}^{(b)}\}$ occurs in C_i, its black edges $\{T_{\mathrm{AH}}^{(b)}, t_b^+\}$ and $\{h_b^+, H_{\mathrm{AH}}^{(b)}\}$ lie in α_i and are also cut, exposing t_b^+ and h_b^+ for insertion.

On the remaining exposed block extremities, $U_i \setminus (D_i \cup G_i)$, where $D_i = \bigcup \delta_i$ and $G_i = \bigcup \gamma_i$, the unique incident gray edges in C_i pair them into a perfect matching. For each $\{x, y\} \in \delta_i$, the gray edges $\{x, T_{\mathrm{DC}}^{(p)}\}$ and $\{y, H_{\mathrm{DC}}^{(p)}\}$ replace the broken DC edge, modeling deletion. For each

$\{t_b^+, h_b^+\} \in \gamma_i$, the gray edges connect t_b^+ and h_b^+ into their host location as specified by β_i, modeling gain.

Thus the rejoining specified by the gray edges in C_i (i.e., by β_i together with δ_i and γ_i) exactly implements the (k_i, d_i, g_i) rejoining rule, with $d_i = |\delta_i|$ and $g_i = |\gamma_i|$. □

Corollary 1. Under Lemma 2, a single step can be decomposed into per-cycle breaks $\tau_i = (\alpha_i, \beta_i, \gamma_i, \delta_i)$ for alternating cycles C_i, $i \in [m]$. For any $I \subseteq [m]$, the union

$$\tau_I = \Big(\alpha' = \bigcup_{i \in I} \alpha_i,\ \beta' = \bigcup_{i \in I} \beta_i,\ \gamma' = \bigcup_{i \in I} \gamma_i,\ \delta' = \bigcup_{i \in I} \delta_i \Big)$$

is a valid (k', d', g')-break with $k' = \sum_{i \in I} |\alpha_i|$, $d' = \sum_{i \in I} |\delta_i|$, $g' = \sum_{i \in I} |\gamma_i|$.

Proof of corollary 2. Alternating cycles are edge- and extremity- disjoint, so each τ_i acts on a disjoint set. Break feasibility (extremity degrees and paired endpoints for insertions/deletions) holds within each τ_i and is preserved under disjoint union. Parameters add component-wise. □

2.5 Scenario enumeration

Definition 6 (ordered partition)**.** Let $C = \{C_1, \ldots, C_m\}$ be the set of alternating cycles of the EBG. An *ordered partition* of C is a sequence $\mathcal{P} = (P_1, \ldots, P_\ell)$ of nonempty, pairwise disjoint subsets of C such that $\bigcup_{t=1}^{\ell} P_t = C$. Each part P_t represents the set of cycles executed together in the t-th step of the scenario.

Given an ordered partition $\mathcal{P} = (P_1, \ldots, P_\ell)$ we define the corresponding scenario by $T_{\mathcal{P}} = (\tau_{P_1}, \tau_{P_2}, \ldots, \tau_{P_\ell})$. Thus, enumeration over all (k, d, g)-scenarios reduces to enumeration of ordered partitions of the alternating cycles of the EBG.

Theorem 1 (bijection with ISA (k, d, g)-scenarios)**.** *Let A and B be genomes such that B is obtained from A under the ISA, and let $G(A, B)$ be the EBG with alternating cycles $C = \{C_1, \ldots, C_m\}$. Then there is a bijection between ordered partitions of C and (k, d, g)-scenarios transforming A into B, where each part of the ordered partition corresponds to one break acting on the union of the cycles in that part.*

Proof. By Lemma 1, every non-isolated connected component of $G(A, B)$ is a color-alternating cycle. Let $C = \{C_1, \ldots, C_m\}$ be the set of these cycles. By Lemma 2, each cycle C_i defines a break $\tau_i = (\alpha_i, \beta_i, \gamma_i, \delta_i)$. And by Corollary 1, For any nonempty subset $S \subseteq C$, the union

$$\tau_S = \left(\bigcup_{C_i \in S} \alpha_i,\ \bigcup_{C_i \in S} \beta_i,\ \bigcup_{C_i \in S} \gamma_i,\ \bigcup_{C_i \in S} \delta_i \right)$$

is a valid break because distinct cycles are vertex-disjoint.

From ordered partitions to scenarios. Let $\mathcal{P} = (P_1, \ldots, P_\ell)$ be an ordered partition of C. Define $T_{\mathcal{P}} = (\tau_{P_1}, \ldots, \tau_{P_\ell})$. Each P_t is nonempty, and the parts are pairwise disjoint. Hence each τ_{P_t} is a valid break acting on extremities disjoint from the extremities used by all other steps.

Applying τ_{P_t} replaces the black adjacencies in the cycles of P_t by the corresponding gray adjacencies. Since the parts of $\mathcal{P}$ cover all cycles of $G(A, B)$, every black edge of the EBG is removed exactly once and every gray edge is created exactly once. Common adjacencies, which do not appear in the EBG, remain unchanged. Isolated vertices are unaffected. Consequently, the final genome has precisely the adjacency set of B.

From scenarios to ordered partitions. Conversely, let $T = (\sigma_1, \ldots, \sigma_\ell)$ be a (k, d, g)-scenario transforming A into B.

We first show that each step σ_t executes a union of complete EBG cycles. Suppose σ_t cuts a black edge $\{u, v\}$ belonging to some cycle C_i. Since T satisfies the ISA, the extremities u and v cannot participate in another breakpoint in a later step. Therefore the new adjacency created at u during σ_t must already be the final adjacency of u in B. Equivalently, σ_t must create the unique gray edge of C_i incident to u.

Let this gray edge be $\{u, w\}$. In order to create $\{u, w\}$, the endpoint w must also be exposed during the same step. Therefore the unique black edge of C_i incident to w must also be cut in σ_t. Repeating this argument around the alternating cycle forces every black edge of C_i to be cut in σ_t, and every gray edge of C_i to be created in σ_t.

Thus no step can execute a proper nonempty subset of an alternating cycle. Every step of T is a union of complete cycles.

Now define $P_t = \{\, C_i \in C : C_i \text{ is executed in step } \sigma_t \,\}$. Since the ISA forbids breakpoint reuse, the sets P_t are pairwise disjoint. Since T transforms A into B, every EBG cycle must be executed in some step. Hence $\mathcal{P}_T = (P_1, \ldots, P_\ell)$ is an ordered partition of C.

The two constructions are inverse. Starting from an ordered partition $\mathcal{P}$, constructing $T_\mathcal{P}$, and then recording which cycles are executed at each step returns the original ordered partition $\mathcal{P}$.

Conversely, starting from a scenario T, recording the cycles executed in each step gives an ordered partition $\mathcal{P}_T$. Rebuilding the scenario from $\mathcal{P}_T$ gives the same sequence of union breaks, up to the ordering of elements inside the sets $\alpha, \beta, \gamma, \delta$.

Therefore, ordered partitions of the EBG cycles are in bijection with the (k, d, g)-scenarios that transform A into B. Consequently, all such scenarios can be enumerated by considering every ordered partition of C. Algorithm 1 performs this enumeration.

□

Algorithm 1 Enumerate (k, d, g)-scenarios from EBG cycles

Input: The alternating cycles $C = \{C_1, \ldots, C_m\}$ of $G(A, B)$ and their corresponding cycle-induced breaks $\tau_i = (\alpha_i, \beta_i, \gamma_i, \delta_i)$.

Output: All (k, d, g)-scenarios transforming A into B.

foreach *nonempty subset* $S \subseteq C$ **do**

  define

$$\tau_S = \left(\bigcup_{C_i \in S} \alpha_i,\ \bigcup_{C_i \in S} \beta_i,\ \bigcup_{C_i \in S} \gamma_i,\ \bigcup_{C_i \in S} \delta_i \right)$$

end

Function `Extend(`$R, \mathcal{P}$`)`**:**

  if $R = \emptyset$ **then**

    output the ordered partition

$$\mathcal{P} = (P_1, \ldots, P_\ell)$$

    output the corresponding scenario

$$T_\mathcal{P} = (\tau_{P_1}, \ldots, \tau_{P_\ell})$$

    return

  end

  foreach *nonempty subset* $S \subseteq R$ **do**

    `Extend(`$R \setminus S,\ \mathcal{P}||(S)$`)`

  end

`Extend(`$C, ()$`)`

Remark 1. Algorithm 1 recursively constructs all ordered partitions of the cycle set C. At each recursive call, R is the set of cycles that have not yet been assigned to a scenario step, and $\mathcal{P}$ is the ordered sequence of parts chosen so far. Choosing a nonempty subset $S \subseteq R$ assigns all cycles in S to the next step of the scenario. The recursion terminates when all cycles have been assigned.

Proof. The algorithm outputs only ordered partitions of C, because each recursive step appends a nonempty subset S of the remaining cycles and removes S from further consideration. Therefore the subsets output at termination are nonempty, pairwise disjoint, and their union is C.

Conversely, let $\mathcal{P} = (P_1, \ldots, P_\ell)$ be any ordered partition of C. At the first recursive call, the algorithm considers every nonempty subset of C, including P_1. After choosing P_1, the remaining set is $C \setminus P_1$, and the algorithm considers every nonempty subset of this remaining set, including P_2. Continuing in this way, there is a unique recursive branch that chooses $P_1, \ldots, P_\ell$ in order. Hence every ordered partition is output.

By Theorem 1, ordered partitions of C are in bijection with (k, d, g)-scenarios transforming A into B. Therefore the algorithm outputs every such scenario exactly once. □

If (k, d, g) are fixed upper bounds on the size of each allowed step, then the enumeration is restricted to subsets $S \subseteq R$ satisfying

$$|\alpha_S| \leq k, \qquad |\delta_S| \leq d, \qquad |\gamma_S| \leq g.$$

In this bounded version, Algorithm 1 enumerates exactly the scenarios whose every step satisfies the prescribed bounds.

Complexity. Let m be the number of alternating cycles in the EBG. The algorithm enumerates all ordered partitions of the cycle set $C = \{C_1, \ldots, C_m\}$.

The number of ordered partitions of an m-element set is the m-th ordered Bell number, also called the Fubini number,

$$F_m = \sum_{\ell=1}^{m} \ell! \left\{ {m \atop \ell} \right\},$$

where $\left\{ {m \atop \ell} \right\}$ is a Stirling number of the second kind. The term $\left\{ {m \atop \ell} \right\}$ counts the partitions of the m cycles into ℓ nonempty unordered parts, and the factor $\ell!$ counts the possible orders of these parts.

Therefore, the number of scenarios output by the algorithm is exactly F_m. Consequently, the running time is necessarily exponential in m, since the output itself is exponential.

If each scenario is output only as an ordered partition of cycle indices, then the total running time is $O(m \cdot F_m)$ because writing one ordered partition requires $O(m)$ total cycle references. If the algorithm explicitly materializes the corresponding breaks $\tau_{P_1}, \ldots, \tau_{P_\ell}$ then the cost per scenario is proportional to the total size of the cycle edges, giving total time $O(|E(G)| \cdot F_m)$. The recursion depth is at most m, so the working space, excluding the output, is $O(m)$ when storing only cycle indices, or $O(m + |E(G)|)$ if the cycle-induced breaks are stored explicitly.

Asymptotically, $F_m \sim \frac{m!}{2(\ln 2)^{m+1}}$, so exhaustive enumeration is practical only when the number of alternating cycles is moderate.

2.6 A dynamic component-based objective for scenario selection

The scenario enumeration framework above identifies (k, d, g)-scenarios with ordered partitions of the alternating cycles of the EBG. This gives all feasible scenarios, but it does not by itself rank them. In this section, we introduce a dynamic objective for comparing scenarios according to the number of current genome components (chromosomes) affected at each step.

The main point is that component involvement is not necessarily a static property of an EBG cycle. A cycle may involve material originating from several reference chromosomes, but after earlier rearrangements have been applied, all of this material may lie on a different number of intermediate components. Thus, the same cycle can have different apparent complexity depending on when it is executed.

Let $C = \{C_1, \dots, C_m\}$ be the set of alternating cycles of $G(A, B)$, and let $\mathcal{P} = (P_1, \dots, P_\ell)$ be an ordered partition of C. As before, the ordered partition defines the $T_\mathcal{P} = (\tau_{P_1}, \dots, \tau_{P_\ell})$, scenario where each τ_{P_t} is the union break induced by the cycles in P_t.

Define the sequence of intermediate genomes by

$$A^{(0)} = A^+, \qquad A^{(t)} = \tau_{P_t}(A^{(t-1)}), \qquad t = 1, \dots, \ell.$$

Thus $A^{(t)}$ is the genome obtained after applying the first t steps of the scenario.

For an intermediate genome $A^{(t)}$, let $\mathrm{Comp}(A^{(t)})$ denote the set of current genome components. These are the ordinary chromosomes of the evolving genome. Auxiliary DC and AH storage components are not counted as components unless explicitly stated otherwise. In particular, deleted fragments represented by DC nodes and unused latent copies stored by AH nodes are treated as auxiliary objects rather than ordinary chromosomes for the purpose of this objective.

For a step τ_{P_t}, let $V(\tau_{P_t}) = \bigcup_{C_i \in P_t} \bigcup \alpha_i$, be the set of extremities exposed by cutting the black edges of the cycles in P_t. Let $V_{\mathrm{act}}(\tau_{P_t}, A^{(t-1)})$ be the subset of these extremities that belong to components of the intermediate genome $A^{(t-1)}$ (without DC/AH nodes).

Definition 7 (dynamic component involvement)**.** Let $\mathcal{P} = (P_1, \dots, P_\ell)$ be an ordered partition, and let $A^{(t-1)}$ be the intermediate genome immediately before step t. The *dynamic component involvement* of step t is

$$\chi_t(\mathcal{P}) = \left| \left\{ K \in \mathrm{Comp}(A^{(t-1)}) : K \cap V_{\mathrm{act}}(\tau_{P_t}, A^{(t-1)}) \neq \emptyset \right\} \right|.$$

Equivalently, $\chi_t(\mathcal{P})$ is the number of current genome components that contain at least one extremity affected by the break τ_{P_t}.

The vector $\chi(\mathcal{P}) = \big(\chi_1(\mathcal{P}), \dots, \chi_\ell(\mathcal{P})\big)$ is called the *component-involvement profile* of the scenario. This profile is path-dependent: the value of $\chi_t(\mathcal{P})$ is evaluated on the intermediate genome $A^{(t-1)}$, and therefore depends on the previously executed parts $P_1, \dots, P_{t-1}$.

This distinction is important. Suppose, for example, that the initial genome contains three reference chromosomes x, y, z. One cycle may represent a translocation between x and y, another a translocation between y and z, and a third a complex rearrangement involving material from all three reference chromosomes. If the complex rearrangement is evaluated relative to the initial genome, it appears to involve three chromosomes. However, if the two translocation cycles are executed first, the relevant material from x, y, z may already lie on a single intermediate component. In that case, the complex rearrangement has dynamic component involvement equal to one, even though it involves material from three original chromosomes.

Several objective functions can be defined from the component-involvement profile. A natural additive objective is

$$\mathrm{cost}_\Sigma(\mathcal{P}) = \sum_{t=1}^{\ell} \chi_t(\mathcal{P}).$$

This objective favors scenarios whose steps affect few current components in total.

A second natural objective is the maximum component involvement

$$\mathrm{cost}_{\max}(\mathcal{P}) = \max_{1 \le t \le \ell} \chi_t(\mathcal{P}).$$

This objective favors scenarios in which no individual step requires a highly multi-component rearrangement.

One may also combine these objectives. For example, one may first minimize $\mathrm{cost}_{\max}$ and then, among scenarios attaining the minimum maximum involvement, minimize cost_{Σ}. This prioritizes avoiding highly complex individual steps, while still preferring scenarios with low total component involvement.

Definition 8 (component-optimal scenario)**.** Let $\Omega(A, B)$ denote the set of (k, d, g)-scenarios transforming A into B, equivalently the set of ordered partitions of the alternating cycles of $G(A, B)$. Let $f : \Omega(A, B) \to \mathbb{R}$, be an objective function defined on scenarios. A scenario corresponding to an ordered partition $\mathcal{P}^\star$ is *component-optimal* with respect to f if

$$f(\mathcal{P}^\star) = \min_{\mathcal{P} \in \Omega(A,B)} f(\mathcal{P}).$$

Typical choices can be $f = \mathrm{cost}_{\Sigma}$, $f = \mathrm{cost}_{\max}$, or a combination of these objectives.

Remark 2 (static versus dynamic involvement)**.** For a cycle C_i, one can define a static involvement score by counting the number of reference chromosomes containing extremities of C_i. Such a score is independent of the order in which cycles are exeuted. In contrast, the dynamic score $\chi_t(\mathcal{P})$ is evaluated on the intermediate genome immediately before step t. Therefore, the same cycle may have different component involvement in different scenarios.

This objective provides a criterion for selecting among the scenarios enumerated in the previous section. While all ordered partitions of the EBG cycles represent feasible scenarios, they may differ in their dynamic component-involvement profiles. The preferred scenario is then not merely one that transforms A into B, but one that does so through intermediate genomes in which each step affects as few current components as possible.

2.7 Computational hardness of $\mathrm{cost}_{\max}$ component-optimal scenario selection

We now consider the computational problem of selecting, among all feasible (k, d, g)-scenarios, one that minimizes the maximum number of genome components affected by any single step.

Let $C = \{C_1, \ldots, C_m\}$ be the set of alternating cycles of the extended breakpoint graph $G(A, B)$. For each cycle C_i, let $\tau_i = (\alpha_i, \beta_i, \gamma_i, \delta_i)$ denote the corresponding cycle-induced (k_i, d_i, g_i)-break. As before, a scenario is represented by an ordered partition $\mathcal{P} = (P_1, \ldots, P_\ell)$ of C, where the t-th step applies the union break

$$\tau_{P_t} = \left(\bigcup_{C_i \in P_t} \alpha_i, \bigcup_{C_i \in P_t} \beta_i, \bigcup_{C_i \in P_t} \gamma_i, \bigcup_{C_i \in P_t} \delta_i \right).$$

For a scenario $\mathcal{P} = (P_1, \ldots, P_\ell)$, recall that the dynamic component involvement of step t is

$$\chi_t(\mathcal{P}) = \left| \left\{ K \in \mathrm{Comp}(A^{(t-1)}) : K \cap V_{\mathrm{act}}(\tau_{P_t}, A^{(t-1)}) \neq \emptyset \right\} \right|,$$

where $A^{(t-1)}$ is the intermediate genome immediately before step t. The bottleneck component-involvement objective is

$$\mathrm{cost}_{\max}(\mathcal{P}) = \max_{1 \leq t \leq \ell} \chi_t(\mathcal{P}).$$

Definition 9 ($\mathrm{cost}_{\max}$ decision problem)**.** The decision version of component-optimal scenario selection under $\mathrm{cost}_{\max}$ is the following problem.

MinMax-Component-Scenario

Input: The alternating cycles $C = \{C_1, \ldots, C_m\}$ of $G(A, B)$, their cycle-induced breaks $\tau_i = (\alpha_i, \beta_i, \gamma_i, \delta_i)$, optional step bounds k, d, g, and an integer q.

Question: Does there exist an ordered partition $\mathcal{P} = (P_1, \ldots, P_\ell)$ of C such that every step satisfies

$$|\alpha_{P_t}| \leq k, \qquad |\delta_{P_t}| \leq d, \qquad |\gamma_{P_t}| \leq g,$$

and

$$\text{cost}_{\max}(\mathcal{P}) \leq q?$$

Equivalently, is there a feasible scenario in which every step affects at most q current genome components?

If the bounds k, d, g are not imposed, the feasibility constraints above are omitted.
Idea: proving NP-hardness by reduction from Vertex Separation:

Definition 10 (Vertex Separation)**.** Let $G = (V, E)$ be an undirected graph with $V = \{1, \ldots, n\}$. A linear layout of G is a bijection $\pi : V \to \{1, \ldots, n\}$. For each step $t \in \{1, \ldots, n\}$, the vertex separator S_t is defined as:

$$S_t = \{u \in V \mid \pi(u) \leq t \text{ and } \exists w \in N(u) \text{ such that } \pi(w) > t\}.$$

The Vertex Separation problem asks whether there exists a layout π such that

$$\max_{1 \leq t \leq n} |S_t| \leq K$$

.

3 Results

3.1 Application to a chromoplexy event in the CA-0098 genome

We applied the proposed reconstruction framework to a chromoplexy event identified by JaBbA in the CA-0098 head and neck cancer genome (Figure 6). The corresponding extended breakpoint graph decomposed into four nontrivial alternating cycles, denoted by

$$\mathcal{T} = \{T_1, T_2, T_3, T_4\}.$$

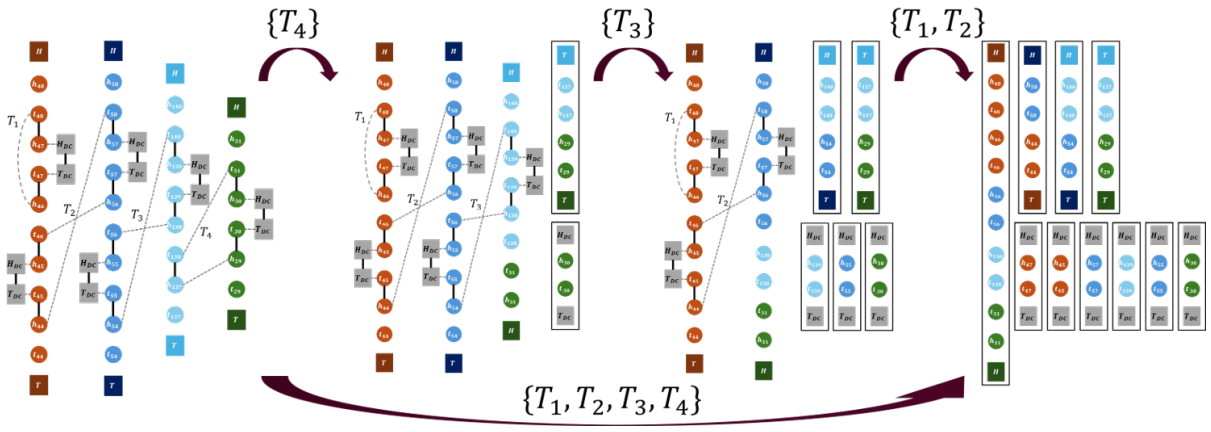


Figure 6: Enter Caption

Under the infinite-sites assumption, each cycle defines an atomic cycle-induced break that must be executed in its entirety within a single rearrangement step. Since the four cycles are vertex-disjoint, any nonempty subset of them may be combined into one valid (k, d, g)-break. Consequently, every feasible reconstruction scenario is represented by an ordered partition of the cycle set $\mathcal{T}$.

For $m = 4$ cycles, the number of ordered partitions is the fourth ordered Bell number,

$$F_4 = \sum_{\ell=1}^{4} \ell! \left\{ {4 \atop \ell} \right\} = 75.$$

Accordingly, the CA-0098 event admits 75 distinct reconstruction scenarios. These scenarios range from a single-step reconstruction,

$$\big(\{T_1, T_2, T_3, T_4\}\big),$$

in which all four cycles are executed simultaneously, to four-step reconstructions in which the cycles are executed individually in one of the 4! possible orders. Intermediate possibilities correspond to scenarios in which selected cycles are grouped into the same rearrangement step.

Although all 75 scenarios produce the same final genome, they differ in the structure of their intermediate genomes and, therefore, in the number of current genome components affected by each step. We evaluated each scenario using the dynamic maximum component-involvement objective,

$$C_{\max}(P) = \max_{1 \le t \le \ell} \chi_t(P),$$

where $\chi_t(P)$ denotes the number of genome components containing at least one extremity affected by the t-th step of scenario P.

The minimum value of the objective was attained by a three-step scenario

$$P^\star = \big(\{T_4\}, \{T_3\}, \{T_1, T_2\}\big).$$

In this reconstruction, cycle T_4 is executed first, followed by cycle T_3, after which cycles T_1 and T_2 are executed jointly. The first two rearrangements alter the component structure of the intermediate genome, such that the final joint step acts on only two current genome components. As a result,

$$C_{\max}(P^\star) = 2.$$

In contrast, the single-step scenario

$$P_{\text{joint}} = \big(\{T_1, T_2, T_3, T_4\}\big)$$

executes all four cycles simultaneously on the initial genome. This joint break affects four genome components and therefore satisfies

$$C_{\max}(P_{\text{joint}}) = 4.$$

Thus, although $P^\star$ and P_{joint} induce the same final set of adjacencies, the optimal sequential reconstruction reduces the maximum number of components involved in any individual rearrangement from four to two.

This case study demonstrates that rearrangement complexity cannot, in general, be determined solely from the final genome graph or from the reference-chromosome composition of individual cycles. The execution of an earlier cycle may merge or reorganize genome components, thereby reducing the number of components involved in subsequent events. The dynamic objective accounts for these changes by evaluating every rearrangement relative to the intermediate genome on which it acts. In the CA-0098 example, this distinction identifies a lower-complexity sequential explanation of the observed chromoplexy structure that is not apparent when all four cycles are interpreted as a single simultaneous event.

4 Discussion

5 Future work

6 Acknowledgements

References